

A TWO-SPACE AUDIT OF SELF-SUPERVISED LEARNING

Does an SSL method keep its promise where we **actually** use it?

Every method enforces its goal in the **loss space**. We deploy from the **backbone**. We measured the gap — then turned the audit into a way to rank and train.

7 loss families, one matched frame

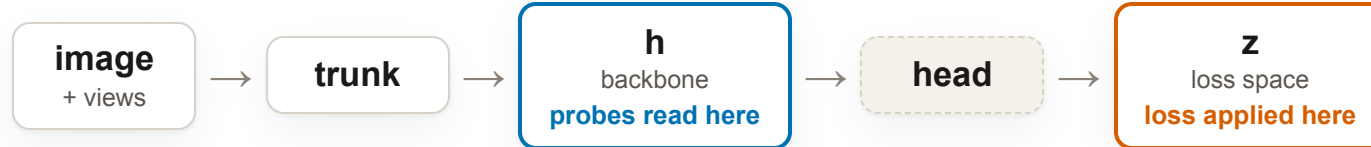
audit → rank → train

Berker Demirel · 2026-07-13

THE QUESTION

The promise is made at **z**. We deploy from **h**.

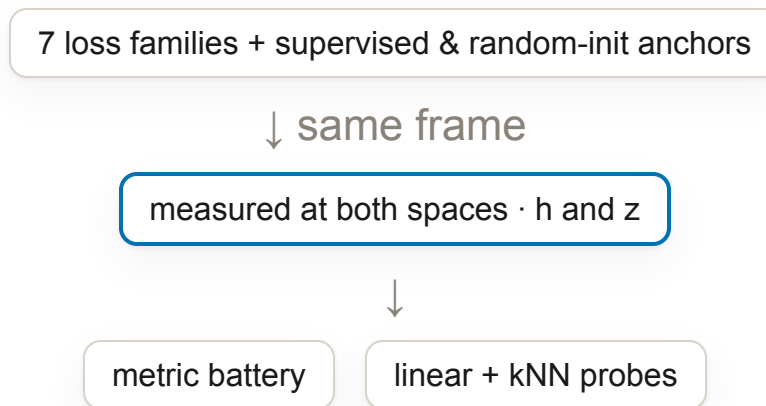
Methods enforce their goal at the space the loss touches (**z**). But probes, kNN and transfer all read the backbone (**h**).



Does the goal survive the head, back to **h** — and does it track quality?

One setup, so the methods are comparable

- Seven loss families, trained by us — **same architecture, same 100-epoch budget, one optimizer.**
- Two scales: a toy set, then **ImageNet-100.**
- Two lenses: an in-house **metric battery** and two frozen probes — **linear** and **kNN.**
- Every prediction is **committed before the numbers exist.**



What we measure — and what each meter can be fooled by

● excess kurtosis

$$E[x^4]/\sigma^4 - 3$$

4th-moment deviation from a Gaussian (whose value is 3), along the worst direction. Heavy tails / clusters $\Rightarrow$ large.

● moment-KL

$$\frac{1}{2}(\text{tr } \Sigma + \|\mu\|^2 - d - \log|\Sigma|)$$

Distance of the features' mean & covariance from a unit Gaussian — the calibration meter. $\downarrow$ = calibrated.

● cos-margin

$$\langle \cos \rangle_{\text{pos}} - \langle \cos \rangle_{\text{rand}}$$

Same idea in cosine: how much closer positive pairs are than random ones. $\uparrow$ = invariant.

● variance floor

$$\min \sigma / \text{mean } \sigma \text{ (per-dim)}$$

Scale-free anti-collapse check. $\approx 1 \Rightarrow$ no dead dimensions.

● sliced-Gaussianity

normality of random 1-D slices

Foolable: in high dim it only senses mean & covariance, not shape. Never read alone.

● align_rel

$$\|a-b\|_{\text{pos}}^2 / \|a-b\|_{\text{rand}}^2$$

View-invariance as a **distance ratio** (positive vs random pairs). $\downarrow$ = invariant.

● effective rank

entropy of the singular spectrum

How many dimensions carry variance. Beware: variance-weighted $\neq$ linear separability.

● class-alignment

$$\text{kmeans-NMI} \cdot \text{between/total var}$$

How much of the geometry lines up with labels — the instrument for semantic structure.

Seven ways to build h — and what each pulls on

CONTRASTIVE

SimCLR

Pull positive views together, push negatives apart.

COVARIANCE

VICReg

Variance floor + decorrelation + invariance.

MASKED RECON

MAE

Reconstruct masked pixels. No augmentations.

DISTRIBUTIONAL

LeJEPa

Push features toward an isotropic Gaussian.

NULL

Random-init

Untrained — the floor everything is scored against.

DISTILLATION

BYOL

Predict an EMA-teacher target; no negatives.

SELF-DISTILLATION

DINO

Match sharpened, centered teacher assignments.

LATENT PREDICTION

I-JEPa

Predict masked latents in representation space.

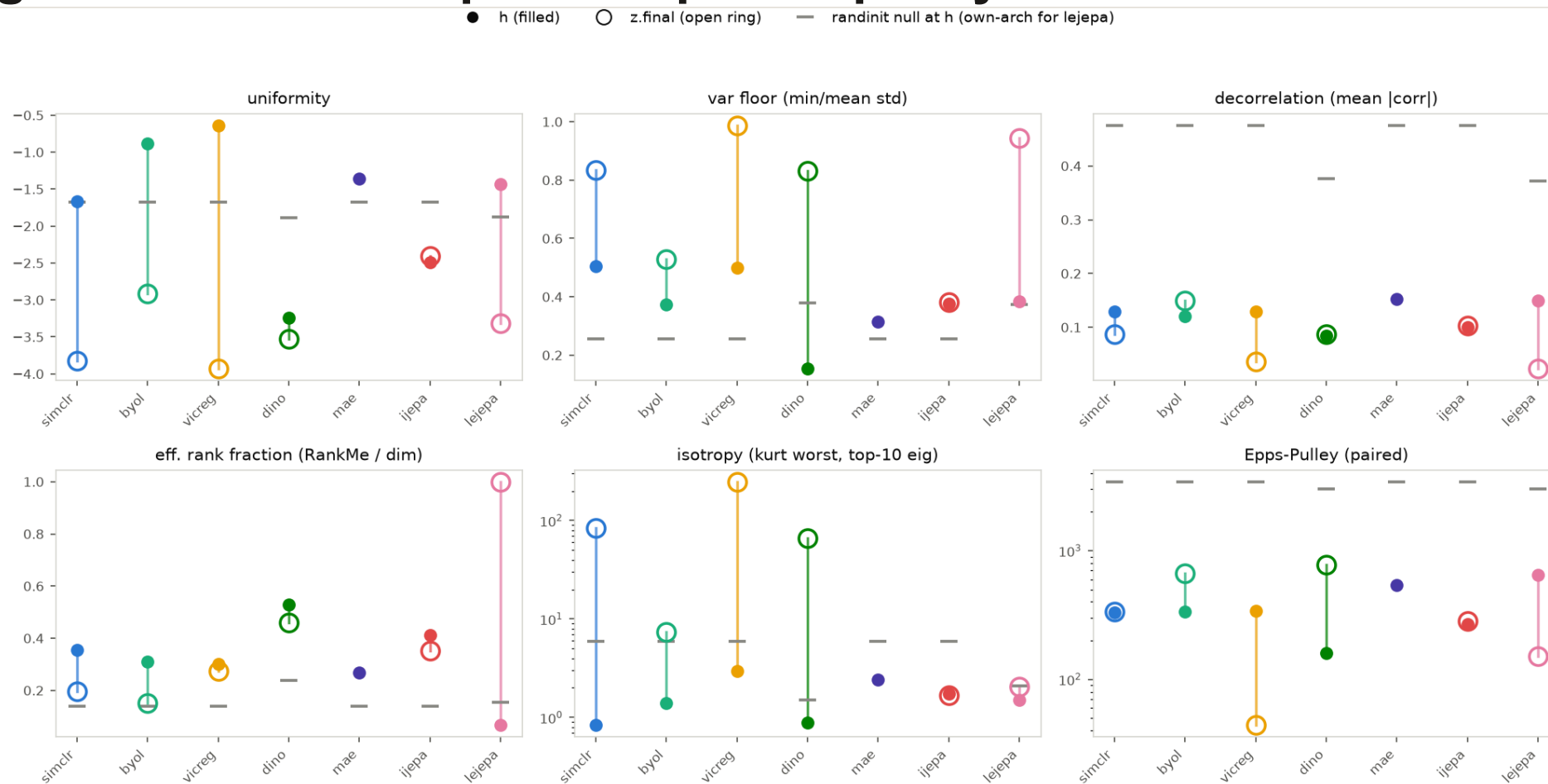
ANCHOR

Supervised

Plain cross-entropy — a reference, not a competitor.

THE OPENING FINDING

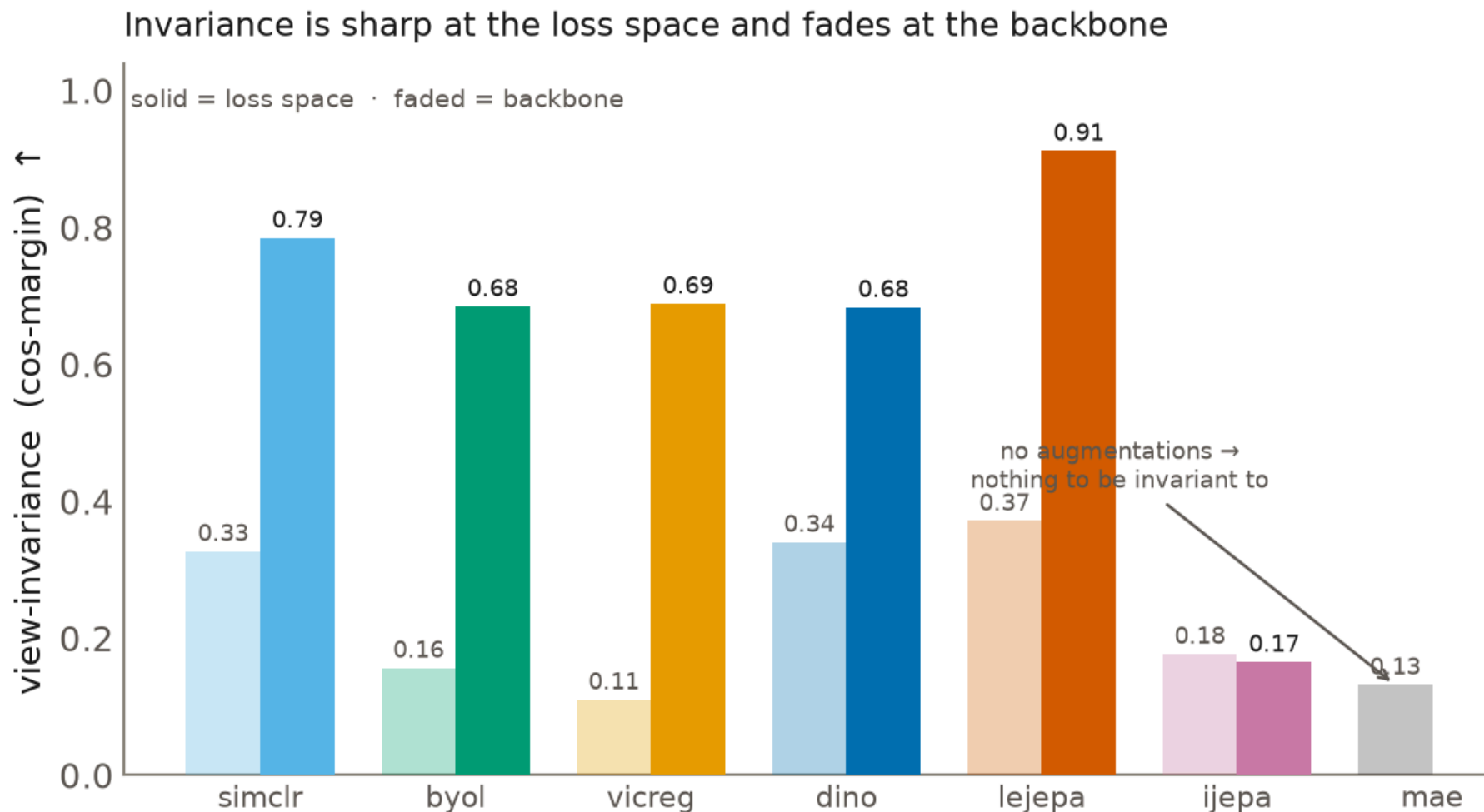
Nothing we measure at h predicts probe quality



Every goal, measured at h (filled) vs z (open ring). Sharp at z, drifting back toward the untrained null at h.

At the loss space, the goal owns the geometry perfectly. At the backbone, no marginal statistic tracks how well the features actually probe. So — is there anything that does?

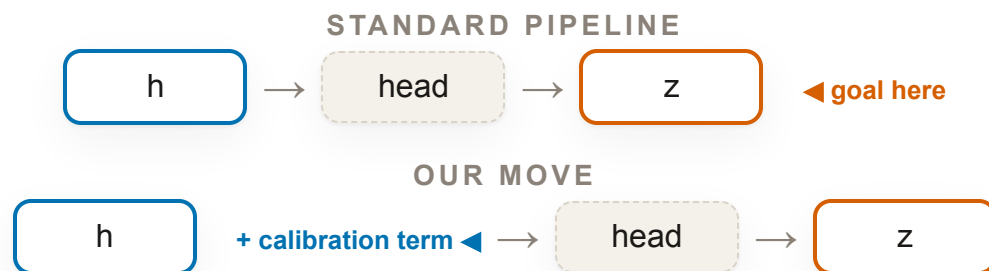
A goal lives only where the loss imposes it



View-invariance, measured as a margin — sharp at the loss space, weak at the backbone.

Each method's goal is enforced at the loss space and fades on the way back to the backbone — the space we actually deploy from.

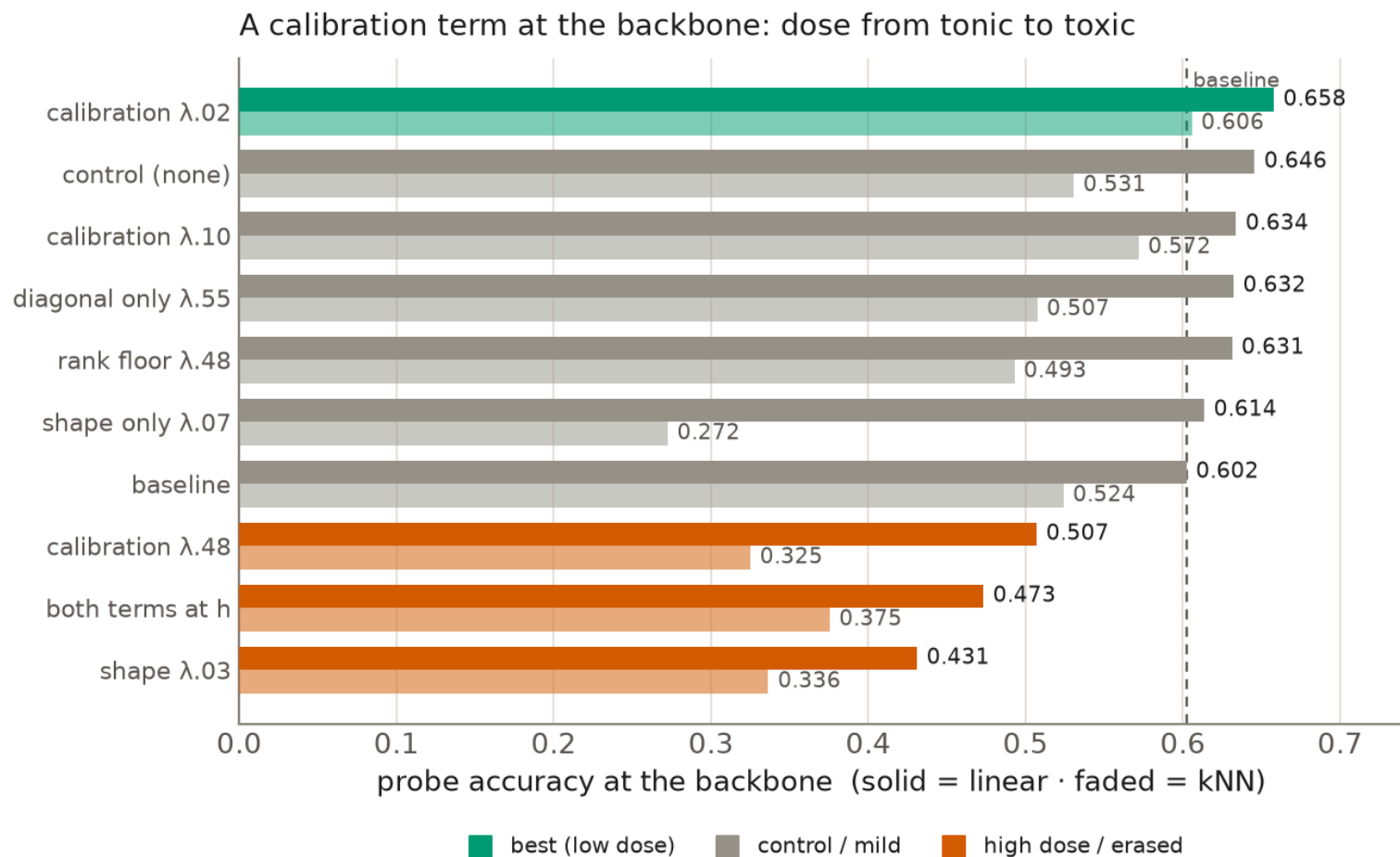
Add a gentle calibration term at the backbone



The bet: the useful part of regularizing the backbone is just **calibrating its mean & covariance**; higher-order shape should be left free for semantics.

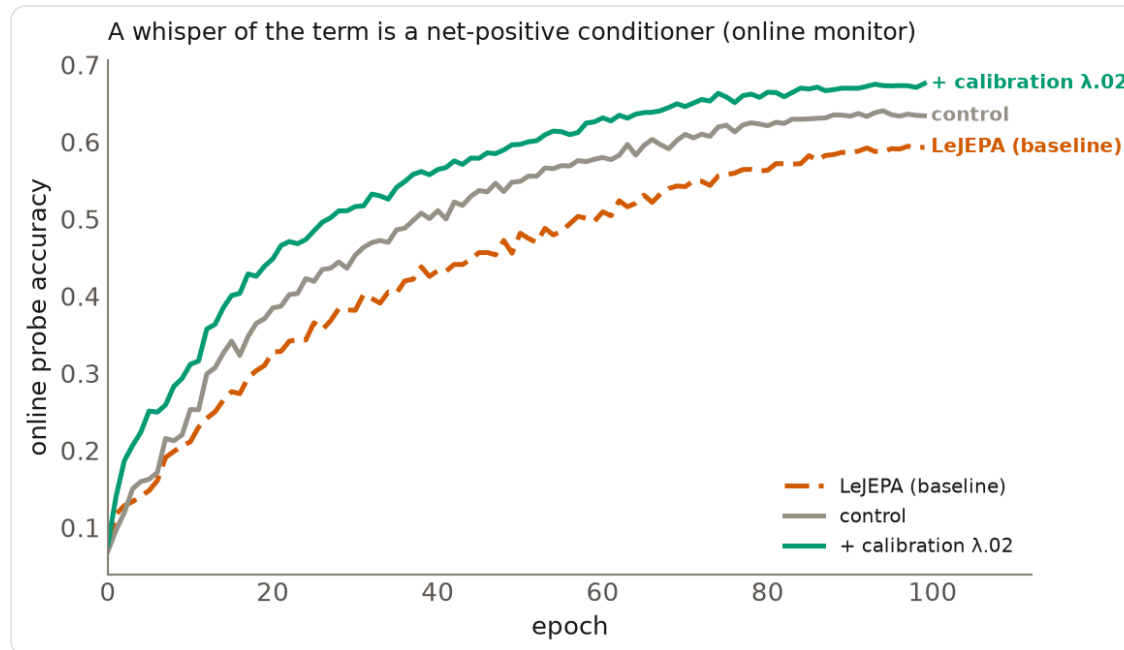
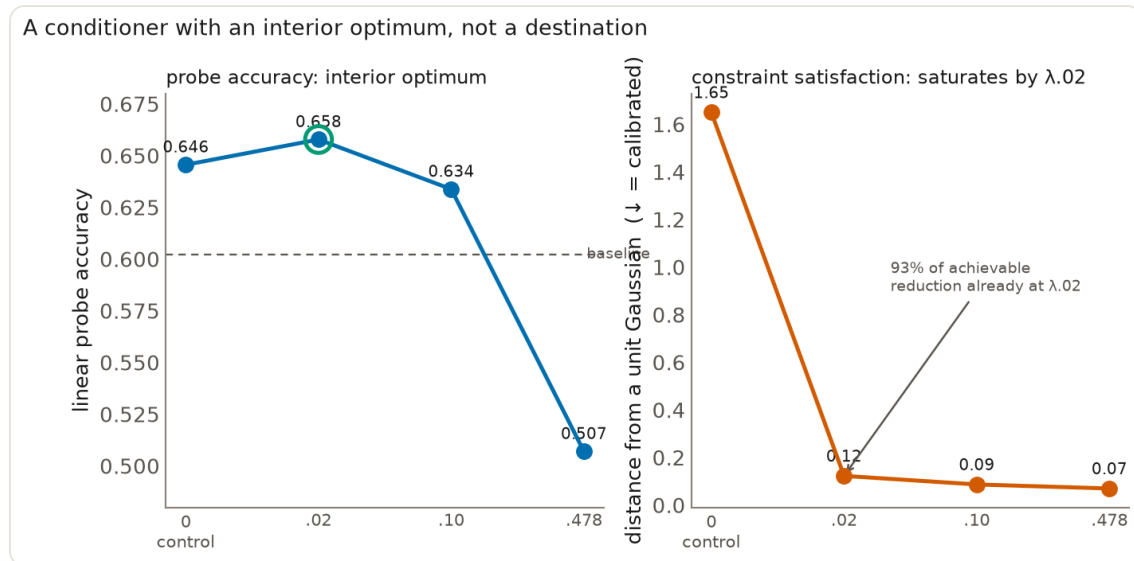
A pure mean+covariance term — blind to clusters by construction. We sweep its strength from a whisper to a shove.

Full calibration erases the semantics



At full strength the term is perfectly enforced — and drags the backbone's accuracy back down toward untrained. Our own first guess, killed: over-calibrating flattens the very structure the features need.

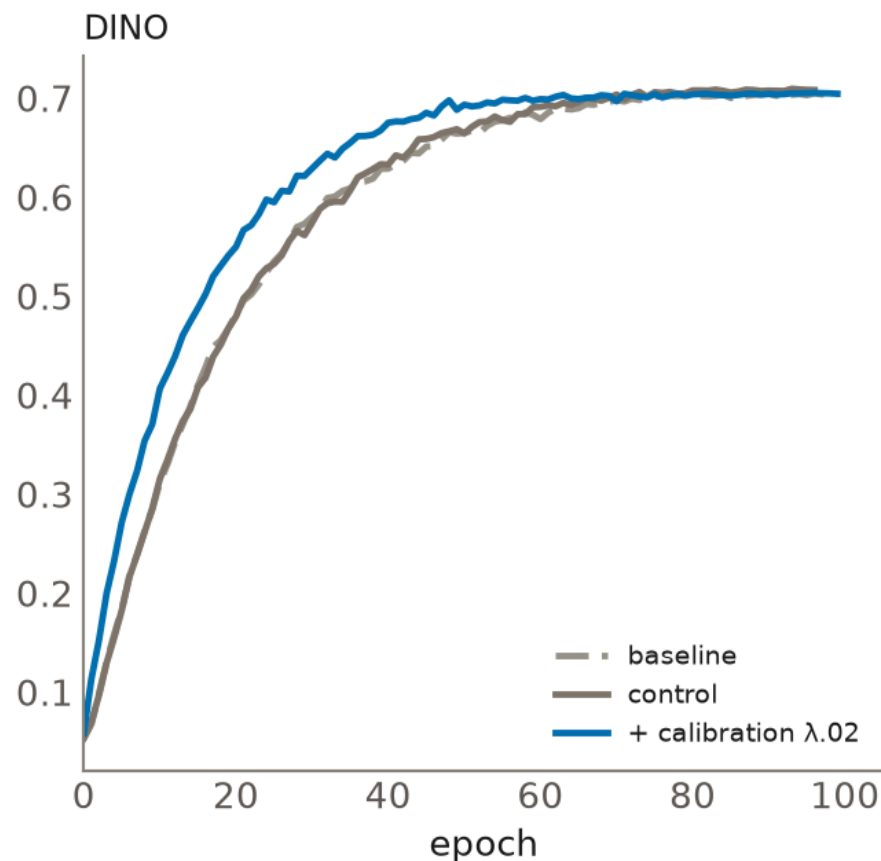
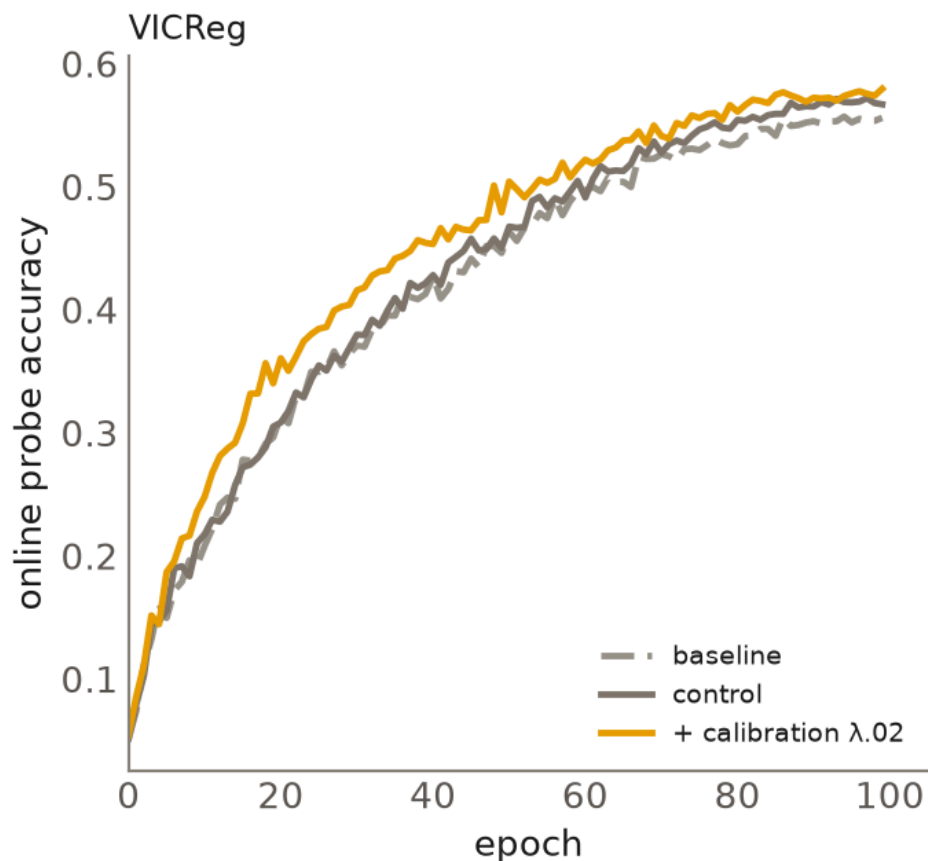
A whisper of the same term helps



At low strength it beats the control on every probe — the best backbone in this family we've produced — and it's genuinely enforced (calibration saturates almost immediately). These terms are conditioners with a sweet spot, not destinations.

The same term on VICReg and DINO

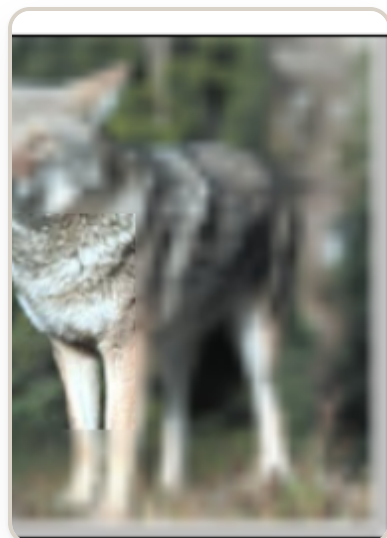
Is the conditioner method-general? (monitor-only — no conclusion yet)



The calibration term makes VICReg better too. For DINO — already the strongest backbone — the improvement is less obvious. (live monitor curves.)

A SHARPER QUESTION

Can we rank models **without labels**?



what the model sees

predict →



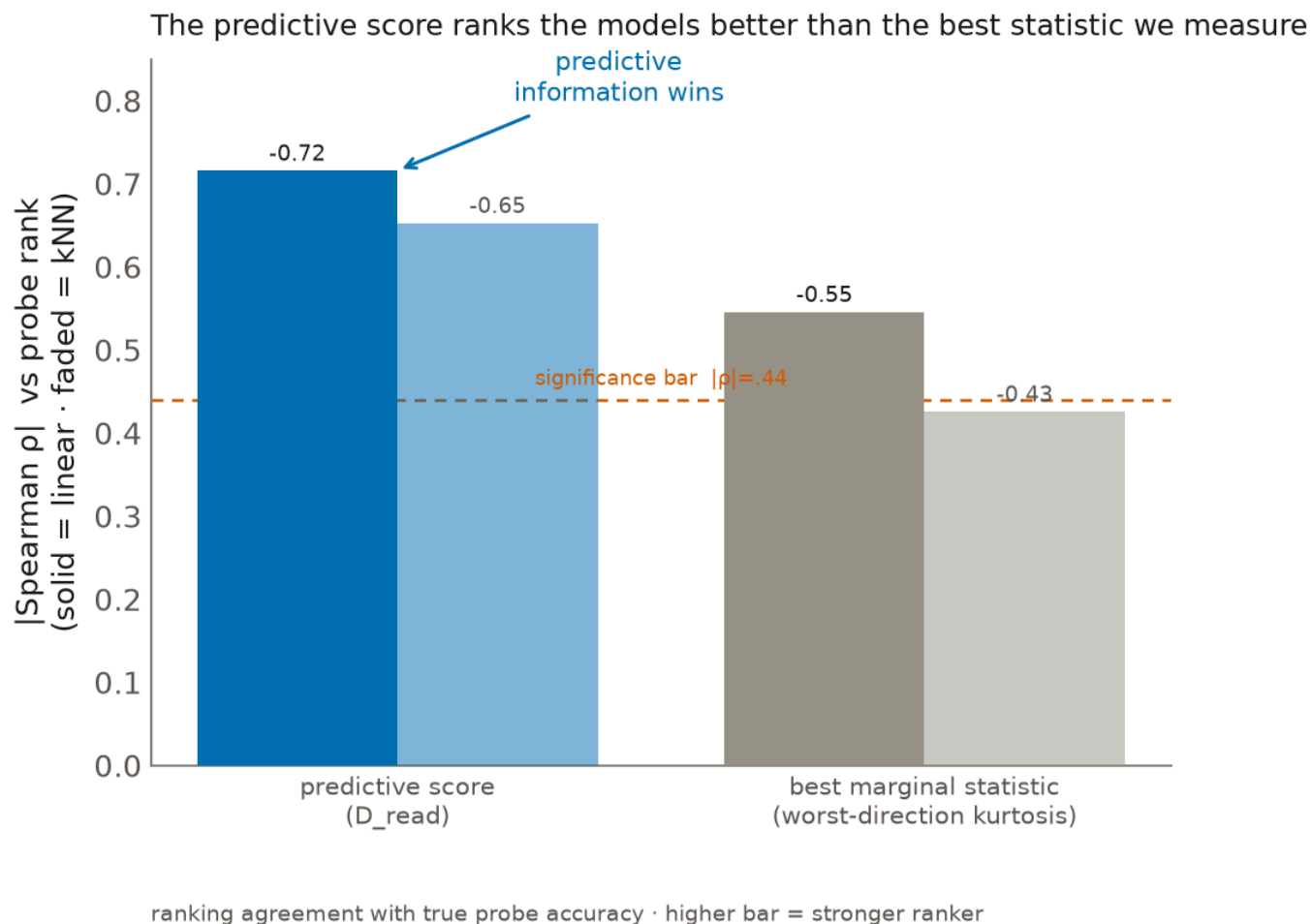
held-out patch → frozen MAE + RFF

Give each model an image; ask how well its features **predict a held-out patch** — encoded as a frozen **MAE** description passed through random Fourier features (**RFF**). How well it recovers that target is the **predictive score, D_{read}** (a held-out linear fit — no labels, no training).

The strongest ranking comes when the target is exactly the frozen MAE tokens of the held-out (non-foveal) region, sketched with RFF — and it's tested on models with the **same architecture and data**, so a win can't be an artifact.

THE RESULT

The predictive score ranks the models — better than anything we measure



Across the whole zoo, the predictive score agrees with true probe accuracy better than every statistic in the battery — and it needs no labels. The one backbone property that tracks quality is how well it predicts what it hasn't seen.

CONCLUSION

What we found

- ✓ Nothing we measure at the backbone predicts how well it actually works.
- ✓ Each method's goal is enforced at the loss space and fades at the backbone.
- ✓ A gentle calibration term at the backbone helps — overdo it and it erases the gains.
- ✓ **A label-free predictive score ranks the models — beating every statistic.**

Where we're headed: use that predictive signal to select and train representations directly.

IN ONE LINE

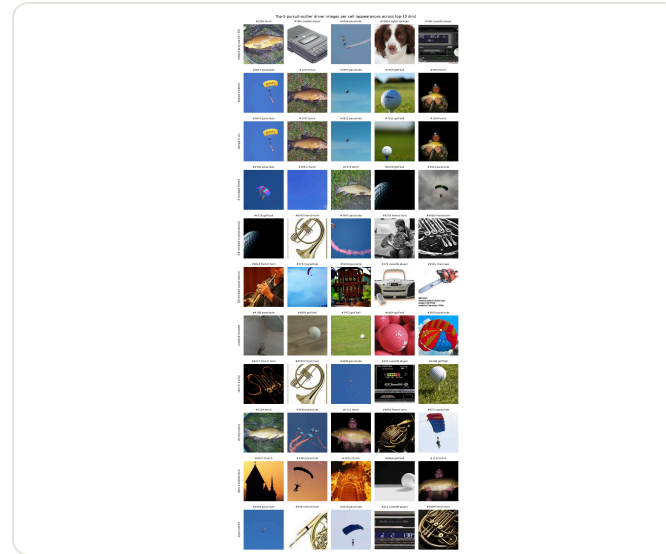
**SSL methods keep their
promise at the loss
space. At the backbone
it's a **balance** — and
what ranks it is
predictive **information**,
not the statistics we
usually measure.**

audit → intervene → rank

every claim tied to a figure

predictions committed up front

Making the "honesty" checks quantitative

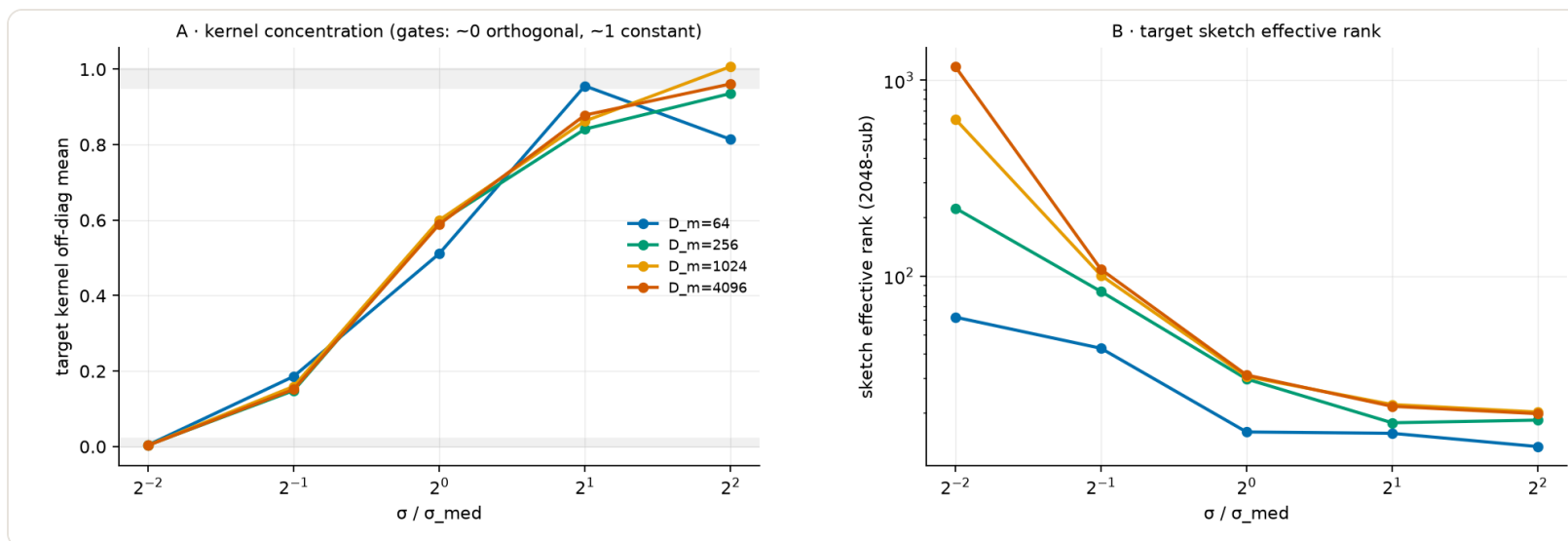


Images behind the structured directions the count picks up.

- A **structured-direction count** that turns "is this Gaussian?" into a number.
- Every isotropy check reports **three reference points** — untrained, covariance-matched, and unit-Gaussian.
- Detection power is calibrated before any such claim leaves the project.

How the predictive score is computed

- Turn the frozen context descriptor into random features (a kernel sketch).
- **D_read** = how well a fixed-capacity linear map predicts it from the backbone, held out.
- **Fix the readout capacity for every model** — capacity differences were pure noise.
- Audit the target first — spectrum, concentration, and how label-aligned it is.



The target is only weakly label-aligned — far from a distillation shortcut.

Why the conclusions are trustworthy

PREDICT FIRST

Every directional prediction is written down before the numbers exist.

NUMBERS LAND RAW

No claim is adopted until it's jointly agreed.

JOINTLY AGREED

A takeaway is adopted only after both of us sign off on it — not when the number looks good.

RE-DERIVE, DON'T PATCH

When a design hits a wall, we rebuild it from first principles rather than tune around it.